

WASP-10b

R-1942

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_150721 · created 2026-08-06T14:46:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457224.8185 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1552 +/- 0.008
Transit depth (Rp/Rs)^2	2.41 +/- 0.25 [%]
Orbital Inclination (inc)	89.68 +/- 0.6
Airmass coefficient 1 (a1)	1.159 +/- 0.012
Airmass coefficient 2 (a2)	-0.1277 +/- 0.0037
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	None
Optimal Aperture	3.19
Optimal Annulus	11.12
Transit Duration (day)	0.0943 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1552 ± 0.008 vs 0.1592 (deviation 0.43σ)
Duration fit vs published	0.0943 d vs 0.0946 d (deviation 0.22σ)
Mid-time O–C	2.2 min
Depth SNR	16.8
Residual scatter	1.01 %

Light curve

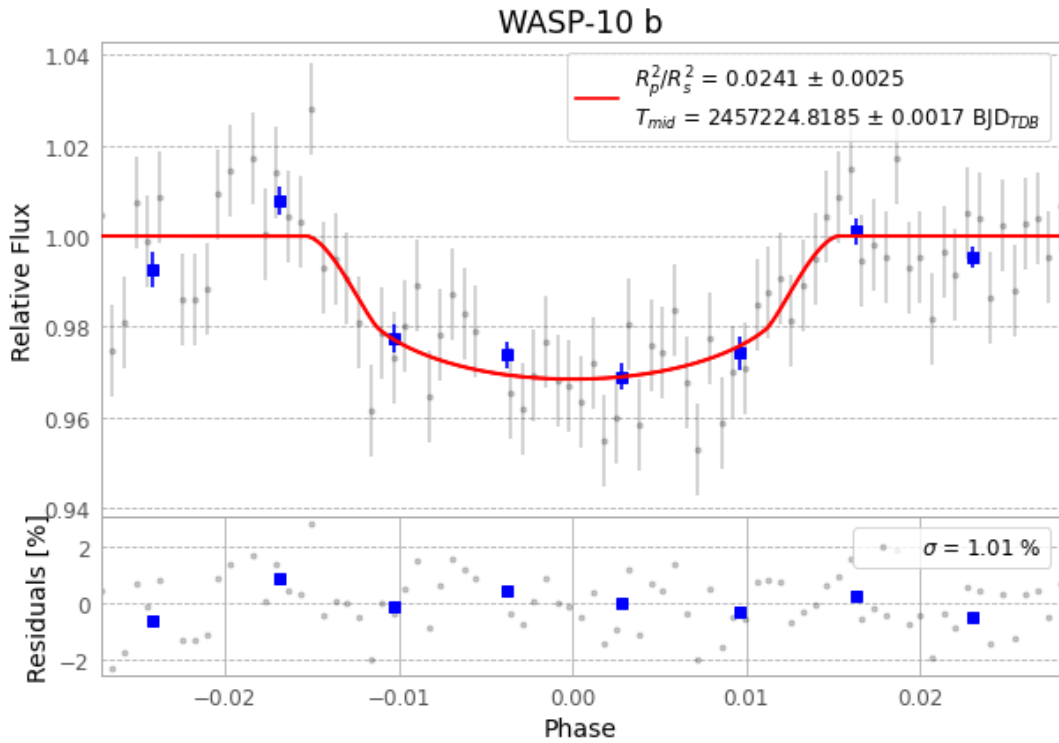


Figure 1 — FinalLightCurve_WASP-10 b_2015-07-20.png (SHA-256 6c6b924fa871711d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [268, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7c7017c4fff37fd9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

83 FITS frames (55 MB), WASP-10150721053312.FITS → WASP-10150721093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-150721.log (a3aaed43f137...), FinalLightCurve_WASP-10 b_2015-07-20.png (6c6b924fa871...), FinalLightCurve_WASP-10 b_2015-07-20.csv (840376231f6f...)

Compute resources

Wall time 182.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 14:46Z.